

Maddies at GroundLM 2026 Shared Tasks: Answer-Preserving Visual Re-grounding for Multi-View Driving VQA

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Abstract

GoldenViewVQA asks a multimodal model to answer a driving-scene question and identify the synchronized camera view that supports its answer. Our first system, QAVEL, used option-blind routing, local image inspection, pairwise view selection, and a final verifier. Tests with GPT-5.6 Luna and Terra showed a recurring problem: extra stages often replaced correct direct predictions, and higher reasoning effort did not repair the cascade. We therefore built Joint Answer-preserving Visual Re-grounding (JAVR) around a GPT-5.6 Sol joint prediction. JAVR freezes the answer, aligns the queried actor or condition across all six views, revisits its disputed cameras with target-specific crops, and changes only the view when a restricted comparator certifies an exact, direct, and complete evidence source. On the 55-item development set, JAVR produced a descriptive one-item increase over its Sol anchor, from 46 to 47 jointly correct predictions, without changing any answer. Our official 59-item test submission reached 79.66% joint accuracy, 81.36% view accuracy, and 94.92% answer accuracy.

1 Introduction

A correct answer can hide a grounding error. In a six-camera driving scene, a model may choose the right option while pointing to a view that does not show the relevant pedestrian, vehicle, or road relation. GoldenViewVQA makes that failure observable: a prediction counts as jointly correct only when both the answer and its supporting camera match the reference (Wang et al., 2026).

We entered the task with a decomposition-heavy system. QAVEL first routed the question without seeing the answer options, inspected promising cameras at multiple scales, and then reconciled this route with a direct joint prediction. The idea was sensible, but the implementation gave too many stages authority to rewrite the final

pair. Across Luna-high, Luna-xhigh, and Terra-high, the full pipeline usually trailed its own direct branch.

That result changed the design question. Instead of asking how to reason more, we asked how to correct the evidence source without disturbing an answer that was already likely to be right. Our submitted system, JAVR, starts from a strong Sol joint prediction and freezes its answer. It then performs explicit cross-view identity matching and permits a view-only override under a strict visual certificate. This restrained controller produced our best held-out result: 47 of 59 test questions jointly correct.

The paper contributes an empirical account of a failed multi-stage design, an inference-only answer-preserving re-grounding method, and an error analysis showing that view attribution remains the dominant bottleneck even at 94.92% answer accuracy. Code, prompts, schemas, and the submitted output are released with the paper.

2 Task and Data

Each item contains a question, four answer choices, and six synchronized nuScenes keyframes: front, front-left, front-right, back, back-left, and back-right (Caesar et al., 2020). The system predicts an answer ID and one supporting view, with NONE_OF_THE_ABOVE available when no camera is sufficient. Joint accuracy is the primary metric.

The shared-task release contains 55 labeled development questions and 59 input-only test questions. The development set has 27 causality, 15 counterfactual, and 13 intent-prediction items. We used the development labels for prompt and controller design; no task-specific training or fine-tuning was performed.

The release stores nuScenes-relative paths rather than images. We pinned dataset revi-

Model	Output	View	Answer	Joint
Luna-high	Direct	.891	.945	.891
	QAVEL	.818	.945	.800
Luna-xhigh	Direct	.855	.945	.855
	QAVEL	.836	.945	.818
Terra-high	Direct	.818	.927	.800
	QAVEL	.836	.891	.782

Table 1: Development accuracy on 55 questions. The full cascade trails its direct branch for every profile. Differences are descriptive; this small sample does not support significance claims.

sion c9402cd4 and validated every path against nuScenes v1.0-trainval metadata. A selective extractor streamed the required keyframe archives and retained 438 unique JPEGs for both splits, rather than the complete sensor release. It downloaded no lidar, radar, sweeps, or maps. Scene and frame overlap between splits is allowed by the task; our system did not use a label-derived frame-memory lookup.

3 From QAVEL to a Smaller Controller

3.1 QAVEL

QAVEL combined an option-blind all-view scout with a direct joint branch. The scout ranked cameras using visible, question-relevant evidence. The two leading cameras were re-examined through a full frame and four overlapping tiles, after which a tournament selected a view. A selected-view pass answered the question, and an adversarial verifier handled disagreements. Causality, counterfactual, and intent prompts used separate evidence contracts.

This design followed two useful intuitions from recent work. Query-conditioned localization can direct visual processing toward relevant regions (Guo et al., 2025), while explicit correspondence is needed because MLLMs often mismatch objects across views (Yeh et al., 2026). QAVEL nevertheless left the answer and view mutable too late in the pipeline.

3.2 What the Model Sweep Told Us

Table 1 rules out the simplest explanation. Increasing Luna’s reasoning effort changed only one joint decision in the full pipeline, and moving to Terra did not improve it. The direct branch remained stronger. A brief attempt to reduce the final verifier’s authority improved development accuracy but regressed on held-out evaluation, so we did not

retain it.

The comparison also cautions against a model-only story. Sol’s later anchor was stronger than Terra’s direct result (46 versus 44 joint items), but Luna-high direct reached 49. Since the prompts and controller also changed for Sol, the final improvement cannot be attributed to model size alone. We chose the GPT-5.6 family because one API exposed image input, configurable reasoning effort, and structured outputs across cost and capability tiers; official documentation positions Luna for high-volume use, Terra as a balance, and Sol as the frontier tier (OpenAI, 2026).

4 JAVR

Figure 1 contrasts the two systems. JAVR is an inference-time controller rather than a trainable neural architecture.

Joint anchor. GPT-5.6 Sol receives the question, options, and all six labeled images. It predicts the view and answer together under a strict JSON schema. The prompt asks for the exact named actor, event, or changed condition before selecting the most direct evidence source. This mirrors the benchmark’s joint protocol and avoids assigning evidence after an answer has been generated.

Answer-frozen localization. The second call receives the question, all views, and the anchor answer, but not the anchor view. The answer is immutable. For every camera, the model records whether the exact target is present, whether the question-critical relation is directly visible, whether target-and-relation coverage is complete, and a normalized target box. This stage addresses the cross-view object mismatch reported by Yeh et al. (2026).

Targeted revisit. If the localizer and anchor disagree, we create a padded crop around the predicted target in each disputed camera. Invalid boxes fall back to deterministic overlapping tiles. A restricted comparator sees only the two full frames, their new crops, the question, and the frozen answer. Candidate order is shuffled deterministically by question ID, and earlier rationales are hidden. This follows the observation that revisiting helps when it introduces complementary visual evidence; simply reasoning again over unchanged evidence may reinforce the first mistake (Ma et al., 2026).

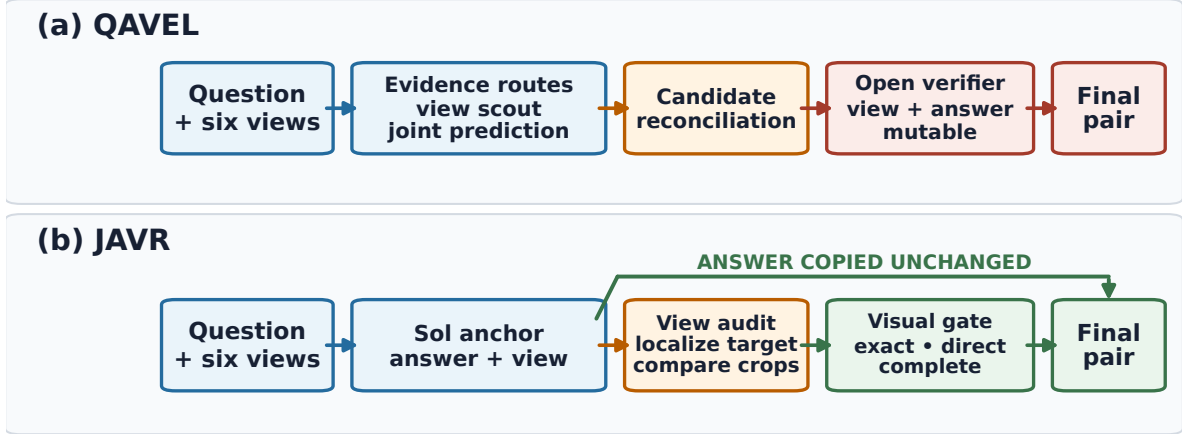


Figure 1: The exploratory QAVEL cascade (top) could replace both decisions at several stages. JAVR (bottom) freezes the Sol answer and gives a narrow, visually certified controller authority over the view only.

Deterministic override. The challenger replaces the anchor view only when the comparator marks it as an exact identity match with a directly visible relation and complete coverage, while finding a material defect in the anchor. An unresolved comparison keeps the anchor. A separate all-view check guards `NONE_OF_THE_ABOVE`. The final answer is asserted equal to the anchor answer in code.

Prompt and schema discipline. The prompts avoid scalar confidence, which proved difficult to calibrate on 55 examples. Instead, each camera receives three categorical checks: target match (exact, possible, different_actor, or absent), relation visibility (direct, partial, or absent), and coverage (complete, partial, or absent). Visible-fact fields are limited to one observable sentence. Local validation requires exactly six entries in canonical camera order, rejects a target box when the target is marked invisible, and checks that the selected view agrees with the unsupported flag. This turns the language model’s intermediate output into evidence for a small deterministic policy rather than a free-form vote.

5 Experiments and Results

5.1 Setup

We used Azure OpenAI’s Responses v1 API. Luna and Terra were evaluated at high reasoning effort, with an additional xhigh Luna run; the submitted JAVR system used Sol-high. Images used the API’s high-detail setting. Every stage had a strict JSON schema followed by local Draft 2020-12 and cross-field validation. Calls were cached and

Scope	Stage	Chg.	View	Answer	Joint
Dev	Sol anchor	0	.873	.909	.836
Dev	Localizer, unguarded	3	.891	.909	.855
Dev	Crop comparison	1	.891	.909	.855
Dev	Full JAVR	1	.891	.909	.855
Test	Full JAVR	2	.814	.949	.797

Table 2: Development-only stage ablation from 55 cached traces and official JAVR test result ($N = 59$, submission 8aa64a419c45). “Chg.” counts views changed from the Sol anchor.

resumable, and concurrency was bounded. The Sol test run used 120 API calls, 944,622 input tokens, 145,052 output tokens (113,447 reasoning), and 500.6 seconds of recorded wall time with concurrency six. No model checkpoint, gradient update, or synthetic training data was used.

For reproducibility, the release records the API model IDs, reasoning settings, prompts, schemas, deterministic shuffle rule, image-crop code, and final predictions. It also includes a selective nuScenes downloader but not the licensed images themselves. Since the models are API-served, there are no redistributable weights; model documentation and the exact deployment identifiers are supplied in place of a checkpoint. The result therefore reproduces the inference procedure and output contract, while exact bitwise replay remains outside our control.

5.2 Development and Official Test Results

Table 2 reconstructs each stage from cached development traces, with no new model calls. Applying the answer-frozen localizer directly changed three views and reached 47/55 joint. The crop com-

parator left two disputes unresolved and retained only the change to `sfall_0135_causality`; the deterministic certificate produced the same final prediction. The three post-anchor rows therefore have identical aggregate accuracy. This ablation supports the controller’s restraint, not a claim that each stage independently improves accuracy. The 46-to-47 difference is one item on $N = 55$ (1.82 percentage points) and is not statistically significant evidence of a general gain.

The official test submission scored 47/59 (79.66%) joint accuracy, with 48/59 views and 56/59 answers correct. These aggregates imply 47 items with both decisions correct, one with only the view correct, nine with only the answer correct, and two with both wrong. Thus, nine test answers were right but attributed to the wrong camera. At $N = 59$, a single item is 1.69 percentage points; nearby scores are not evidence of statistical separation.

The label-free audit found two test disagreements and overrides (2/59, 3.39%), versus three disagreements and one override on development (1/55, 1.82%). Both test challengers passed target, relation, and coverage checks; their anchors passed these checks in 2/2, 1/2, and 0/2 cases. Coverage therefore drove both overrides, with relation visibility additionally decisive once. Only one of three development challengers passed all checks. These sparse rates are descriptive.

The 81.36% micro versus 59.30% macro test view accuracy suggests uneven class recall. Two altered views cannot constitute class balancing, and hidden labels prevent assigning the gap to particular cameras. The audit explains controller sparsity, not the hidden confusion matrix.

5.3 Remaining Failures

Intent prediction remained the weakest development group at 8/13 joint, compared with 26/27 for causality and 13/15 for counterfactual questions. A still image can show orientation or lane commitment, but it cannot always establish future motion. JAVR also missed all three development examples whose view was `NONE_OF_THE_ABOVE`. A dedicated abstention branch was considered after the main run, but small pilot checks did not improve joint accuracy and we did not pursue it.

Development is dominated by the front camera (38/55 items), where JAVR recovered 37. It recovered 7/8 front-left, 1/2 back-left, and all five examples across the remaining real-view classes, but

none of three unsupported cases. We did not tune a class prior from these sparse counts: that could trade away the front-camera decisions dominating joint accuracy. We therefore report the official test macro score without inferring its hidden per-class confusion matrix.

6 Conclusion

Our main result came from reducing authority, not adding another judge. QAVEL showed that a plausible decomposition can lose correct pairs when routing, answering, and verification errors accumulate. JAVR instead treats a strong joint answer as an anchor and asks a narrower question: can fresh, target-specific visual evidence certify a better source view? That change yielded 79.66% official joint accuracy. The remaining answer-correct/view-wrong cases point to cross-view identity and rare-view calibration as the next problems, while the current evidence does not support a larger cascade.

Limitations

The public development set has only 55 items, and JAVR’s measured gain there is one question. The stage ablation is development-only; the test trace audit is label-free and, by itself, cannot determine whether either override was correct. API-served models may change behind stable deployment names, which limits exact reproducibility; we release prompts, schemas, settings, token accounting, and outputs but cannot redistribute model weights. Our model comparisons are not fully controlled because the Sol move coincided with a controller redesign. The benchmark uses one synchronized timestamp and one reference view, although some real driving decisions need temporal or multi-view evidence. Finally, benchmark performance should not be read as evidence that this system is safe for autonomous driving.

Ethical Considerations

GoldenViewVQA references real nuScenes traffic images. We obtained them through official access and release neither images nor personal information. They are used only for benchmark research, not identification or deployed driving. We disclose all API models and did not train on hidden test labels.

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